

Care of the new born baby



Objectives

At the end of the lesson, students will be able to:

- Demonstrate essential immediate new born care
- Be familiar with new born resuscitation
- Describe low birth weight care

Introduction

- The transition from intrauterine to extra uterine life is a complex process involving virtually every organ system in the body.
- The most dramatic changes are seen in the lung and the cardiovascular system.
- Failure to adequately make this transition can be life-threatening and these infants often require supportive care.

Introduction cont'd

➤ To save the life of thousands of newborns dying every year in Ethiopia, we should mainly focus on the triads of activities that should be carried out routinely:

1. Ensure the provision of Essential newborn care to all babies
2. Early detection and management of common newborn problems particularly Birth asphyxia, and neonatal infection.
3. provision of extra care and creation of an environment of thermal comfort to all preterm and low birth weight babies

Essential Newborn Care

- 90% of newly born babies make the transition from intrauterine to extra-uterine life without difficulty.
- They require little to no assistance to begin spontaneous and regular respirations.
- Approximately 10% of newborns require some assistance to begin breathing at birth and only about 1% need extensive resuscitative measures to survive.

ENC cont'd

❖ **STANDARD PRECAUTIONS**

- Always remember the importance of observing **precautions** to help protect the mother and baby and ourselves from infections with bacteria, viruses including HIV
- **At delivery: WHO five cleans**
 - ✓ Hands of attendants (washed with soap)
 - ✓ Surface for delivery
 - ✓ Cutting instrument for cord(i.e. scissors, blade)
 - ✓ String to tie cord
 - ✓ Cloth to wrap baby and mother (**all should be sterile**)

ENC cont'd

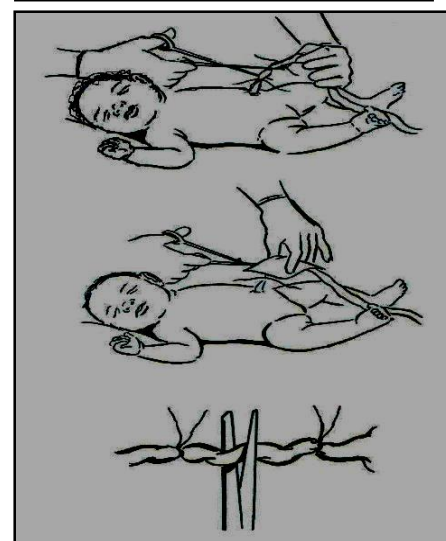
➤ The following are the steps of immediate care which should be given to all babies at birth.

1. Dry and provide warmth(Replacing the wet towels)

- ✓ Deliver the baby in prone position on the mother's abdomen, **face turned to the side.**
- ✓ Dry the newborn thoroughly.
- ✓ Check the baby's breathing while drying.
- ✓ Remove wet cloth.
- ✓ Place the newborn on the mother's abdomen in **skin-to-skin contact.**
- ✓ Cover the back with a dry blanket.

Steps in Immediate Newborn Care After Birth

- **Step 1:** Deliver baby on to mother's abdomen or into her arms
- **Step 2:** Dry baby's body with dry towel. Wipe eyes. Wrap with another dry one and cover head
- **Step 3:** Assess breathing and color. If < 30 breaths per minute, blue tongue, lips or trunk or if gasping then- start resuscitating
- **Step 4:** Tie the cord two fingers from abdomen and—another tie two fingers from the 1st —one (if no clamp) Cut the cord between the 1st and 2nd- tie (clamp)



Immediate Newborn Care After Birth; cont'd

- **Step 5:** Place the baby in skin-to-skin contact and on the breast to initiate breastfeeding
- **Step 6:** Apply Tetracycline eye ointment once
- **Step 7:** Give Vitamin K 1mg IM on anterior mid thigh
- **Step 8:** Weigh baby– (if <1500 gm refer urgently)



General Consideration

- Do your best not to remove **the vernix** (the creamy, white substance which may be on **the skin**) as it protects the skin and may help prevent infection.
- Then wrap the baby with another dry cloth and cover the head.
- Do not suction unless mouth/nose are **obstructed with secretions or other material**
- Do not ventilate unless the baby is **floppy and not breathing**
- Once a baby is born, and while being dried, the baby's breathing should be assessed.

Question ???



What are the key principles and practices in cord care?

Cord Care

❖ Time of clamping

When should the cord be clamped after birth?

- When the cord pulsations stop
- Between **1 and 3** minutes
- Between **30 seconds - 1** minute in **preterm**

A. Early cord clamping

- **Advantages:** The length of the third stage of labor is reduced.

Cord Care cont'd

- **Disadvantages:**

- Preterm babies have an increased incidence of **respiratory distress**.

- ✓ Is associated with lower hematocrit.

- ✓ It increases the risk of **Feto – maternal transfusion – Rh**

Cord Care cont'd

B. Delayed cord clamping

- Delayed for preterm babies and rhesus negative mothers
- **Advantages-**
 - ✓ Preterm babies will benefit from resulting increase in circulating blood, and higher neonatal **hematocrit and hemoglobin** should be obtained

Cord Care cont'd

- **Disadvantages**

- ✓ over loading the **baby's system**

C. Leaving maternal end of the cord unclamped when cord is cut.

➤ Before allowing the maternal end of the cord to remain unclamped, a **second twin** must be excluded.

Cord Care cont'd

■ Advantages

- ✓ There is less blood loss
- ✓ Lower incidence of manual removal of the placenta
- ✓ Rhesus negative mothers carrying rhesus **positive babies** are less likely to develop antibodies

- **Disadvantages** – A second twin may die when it is unclamp

Cord Care cont'd

- If the glove contaminated , change the glove
- Tie the cord securely in two places
 - ✓ Tie the first one **two fingers** away from the baby's abdomen.
 - ✓ Tie the second one **four fingers** away from the baby's abdomen.
 - ✓ Make sure that tie is well secured.
 - ✓ Make sure that the thread you used to tie the cord is clean and safe.

Cord Care cont'd

➤ Cut the cord between the ties

- Use a sterile scissors.
- Use a small piece of **cloth or gauze** to cover the part of the cord you are cutting so no blood splashes on you or on others.
- Be careful not to **cut or injure** the baby.
- Either cut away from the baby or place your hand between the cutting instrument and the baby.

Cord Care cont'd

- Do not put anything on the cord stump
 - Note: observe for oozing blood:
 - If blood oozing, place a second tie between the skin and the first tie.
 - Usually falls **off 4-7 days** after birth
 - Until the cord falls off, place the **cord outside** the nappy to prevent contamination with urine/feces
 - Wash with soap and clean water only (if soiled)

Thermal Protection

❖ Newborn physiology

- Normal temperature: 36.5–37.5°C
- Hypothermia: $< 36.5^{\circ}\text{C}$
- Stabilization period: 1st 6–12 hours after birth
 - ✓ Large surface area
 - ✓ Poor thermal insulation
 - ✓ Small body mass to produce and conserve heat
 - ✓ Inability to change posture or adjust clothing to respond to thermal stress

Thermal Protection cont'd

➤ Increased hypothermia

- ✓ Newborn **left wet while waiting** for delivery of placenta
- ✓ Early bathing of newborn (**within 24 hours**)

Hypothermia Prevention

- Deliver in a warm room
- Dry newborn thoroughly and wrap in dry, warm cloth
- Give to mother as soon as possible
 - ✓ Skin-to-skin contact first few hours after childbirth
 - ✓ Promotes bonding
 - ✓ Enables early breastfeeding
- Check warmth by feeling newborn's feet every 15 minutes
- Bathe after temperature is stable (after 24 hours)

Eye (Care) Prophylaxis

- Immediately after the head is born, the baby's eyes should be cleaned
- The eyes of all infants must be protected against **gonococcal infection** by applying :
 - **1% silver nitrate** drops-the best proven therapy
 - **0.5% erythromycin or 1% tetracycline ointment**
(alternative measures ,this also prevents Chlamydia infections)

Eye (Care) Prophylaxis cont'd

- 2.5% povidone-iodine-may be effective at one time prophylactic agent
- Infants born to mothers with active gonorrhoea are at high risk for developing neonatal conjitivities and should be given:
 - A single 125 mg/im injection of ceftriaxone for prophylaxis
 - For low birth weight infants, the dose is 25-50mg/kg

Vernix

- Is a white, cheesy coating on the skin of new born
- Is a combination of **skin secretions** and **skin cells**.
- Excess Vernix can be cleaned from the baby, it is **not necessary to remove all** of it as it has a protective effect

Meconium

- Is the dark green fecal material that babies pass after delivery and sometimes before delivery.
- If the amniotic **fluid is green colored**, this is due to the presence of meconium
- Babies subjected to **intrauterine stress** often pass meconium before delivery, but so do many non-stressed fetuses.

Meconium cont'd

- Is significant because it indicates the need to thoroughly **clear the airway.**
- In a hospital setting, accomplished with endotracheal visualization or intubation.
- In operational settings, careful suctioning with any available device will be performed

Vitamin K

- Give Vitamin K, 1mg IM on anterior lateral thigh (while baby held by his mother)
- During the first few hours following delivery, a single injection of **0.5 to 1.0 mg** of natural vitamin K.
- Given according to national guidelines ampoule (**1 mg/0.5ml or 1 mg/ml**) IM once (do not use 10 mg/ml ampoule)

vaccine

- oral polio = 0
- hepatitis B Virus if the mother has active HBV infection.
- BCG vaccines
- If sero +ve ART prophylaxis

Skin -to-skin (STS) contact

- Place the baby in skin-to-skin contact and on the breast to initiate breastfeeding
- The warmth of the mother passes easily to the baby and helps stabilize the baby's temperature
 - Put the baby on **the mother's chest**, between the breasts, for skin-to-skin warmth
 - Cover both mother and baby together with a warm cloth or blanket
 - Cover the baby's head

Skin -to-skin (STS) contact cont'd

- The first **skin-to-skin contact** should last uninterrupted for at least **1 hour** after birth or until after the first breastfeed
- The baby should not be bathed at birth because a bath can cool him dangerously.
- After 24 hours, he can have the first **sponge bath**, if his temperature is stabilized

Skin -to-skin (STS) contact cont'd



Early and Exclusive Breastfeeding

- Early contact between mother and newborn
 - ✓ Enables breastfeeding
 - ✓ Rooming-in policies in health facilities prevents nosocomial infection
- Best practices
 - ✓ No prelacteal feeds or other supplement
 - ✓ Giving first breastfeed within one hour of birth
 - ✓ Correct positioning to enable good attachment of the newborn
 - ✓ Breastfeeding on demand
 - ✓ Psycho-social support to breastfeeding mother

Early and Exclusive Breastfeeding (cont'd)

➤ Starting to breastfeed

- ✓ Colostrum is the first milk secreted and is important for the baby for nutrition and disease protection
- ✓ Most babies are ready to feed 15-55 minutes after birth. Success at the first feeding often indicates successful later breastfeeding

➤ Self-attachment

- ✓ Place baby face down on mother's abdomen
- ✓ Support baby as it moves towards breast
- ✓ Allow the baby time to mouth the nipple before taking it into the mouth

Early and Exclusive Breastfeeding (cont'd)

➤ Signs that baby is getting enough milk:

- ✓ The baby passes urine at least 6 times in 24 hours
- ✓ You can hear the baby swallow the feeding
- ✓ The mother's breast feels softer after a feed
- ✓ The baby gains weight over time (after the first week)
- ✓ The baby seems content/satisfied after feeding.

Counseling

- Even if the mother is being discharged a few hours after childbirth, she should be counseled about:
 - ✓ Exclusive breastfeeding
 - ✓ Hygiene – eye and cord care
 - ✓ Thermal protection
 - ✓ Danger signs and what to do about them

Immediate Care of Low Birth Weight Newborn

- Low birth weight babies are those who are born with a weight below 2.5 kg . **VLBWT < 1.5kg**
- Further, classified based on:
 - Birth wt.
 - GA
- Extra care for **warmth(kangaro mother)**
- Extra care **in feeding**

Immediate Care of Low Birth Weight Newborn cont'd

- Extra care **against infection**
- Extra care to detect and evaluate danger signs
- Birth Asphyxia
 - ✓ Immediate Action:
 - ✓ Dry baby and turn him on his side and quickly rub his back to help him start breathing
 - ✓ If necessary, provide resuscitator's (**mouth and cheeks only**) as mouth-to-mouth or mouth-to-nose resuscitation

Immediate Care Of Low Birth Weight Newborn cont'd

- Practice early and **frequent exclusive breastfeeding**
- Keep the baby warm, ideally through practicing **kangaroo mother care**
- Delay **bathing for one week** or until baby is well stabilized
- When bathing, use warm water, dry, wrap well or place in **skin-to skin** contact

Unnecessary Procedures

Not routinely recommended for all neonates

- Routine suctioning
- Early bathing/washing
- Foot printing
- Giving sugar water, formula or other prelacteals and use of bottles and pacifiers
- Application of **alcohol, medicines** and other substances on the **cord stump** and bandaging the cord stump or abdomen

Neonatal Resuscitation

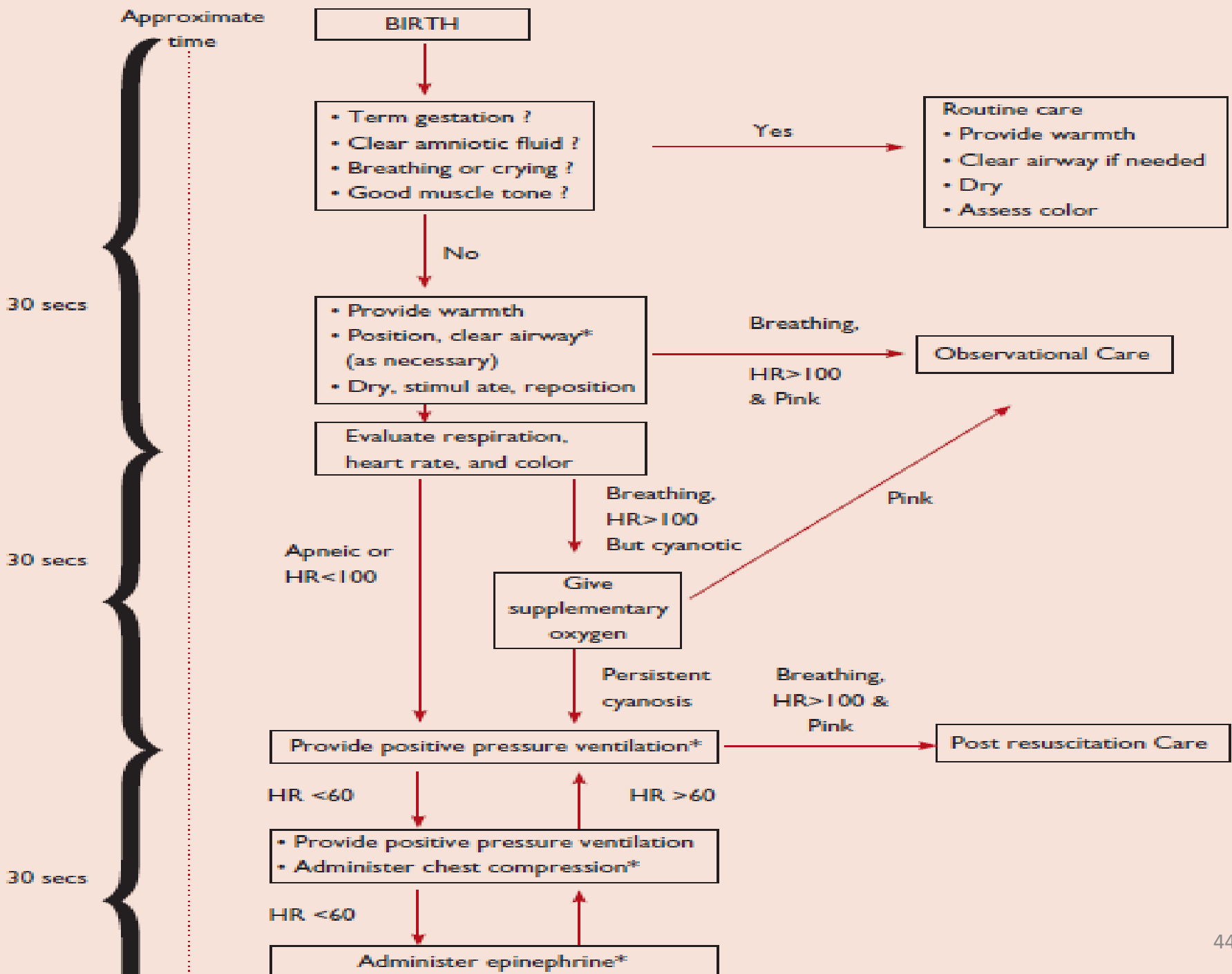
- Transition from IU life to EU (Extra uterine)life is the most important period in neonatology
- Most neonates complete the transition to extra uterine life without difficulty and require only the routine care; but **5 to 10 %** require resuscitation.
- Failure to initiate and maintain effective respiration is the most common delivery room emergency for neonates.

Goals of neonatal resuscitation

- ✓ Prevent the morbidity and mortality associated with hypoxic-ischemic tissue injury.
- ✓ Reestablish adequate spontaneous respiration and cardiac output.
- ✓ Improve the likelihood of preventing brain damage.
 - ❖ In order to achieve this goal High-risk situations should be anticipated by
 - Pregnancy, labor, and delivery history and
 - Identification of signs of fetal distress.

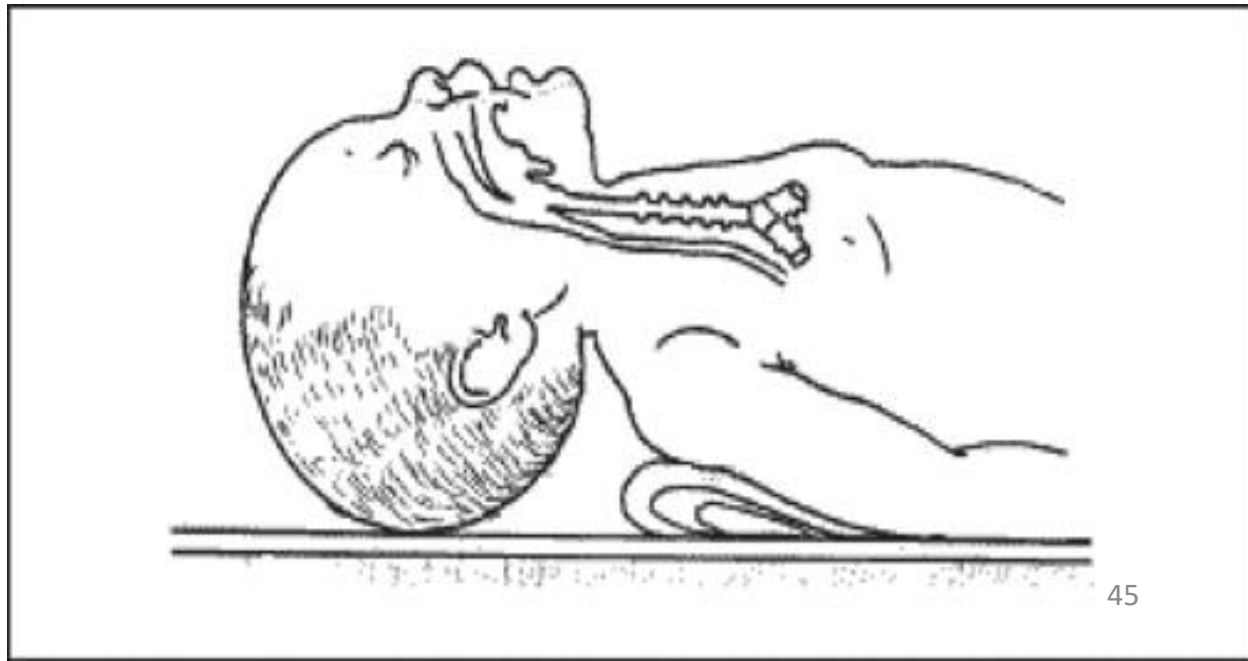
When anticipating resuscitation

- The radiant warmer is turned on and is heating.
- The oxygen source is open with adequate flow through the tubing.
- The suctioning apparatus is tested and is functioning properly.
- The laryngoscope is functional with a bright light.
- Testing of resuscitation bag and mask demonstrates an adequate seal and generation of pressure.



What are the initial steps and how are they administered?

- A. Provide warmth:** The baby should be placed under a radiant warmer
- B. Position; clear airway (as necessary)**
 - Position by slightly extending the neck Dry, stimulate, reposition



Clear airway (as necessary)

- ❖ **In vigorous babies** (a vigorous baby is one who has strong respiratory efforts, heart rate > 100 bpm and a good muscle tone), no suctioning of oro-pharynx or the trachea is required.
- ❖ **In all non-vigorous babies** suctioning of oro-pharynx needs to be done.
- If, such babies were also delivered through meconium stained amniotic fluid, oropharyngeal suction would need to be followed by tracheal intubation to clear it of meconium (if skills are available).

How do you clear the airway?

- Secretions may be removed from the airway by wiping the nose and mouth with a towel or by suctioning with a mucus extractor or suction catheter.
- If the newborn has copious secretions coming from the mouth, turn the head to the side.
- The mouth is suctioned before the nose to ensure that there is nothing for the newborn to aspirate if he or she should gasp when the nose is suctioned.

C. Dry, stimulate, reposition

- Often, positioning the baby and suctioning secretions will provide enough stimulation to initiate breathing.
- Drying will also provide stimulation.
- If 2 people are present, the second person can be drying the baby while the first person is positioning and clearing the airway.
- As part of preparation for resuscitation, you should have several pre-warmed towels available.

What other forms of stimulation may help a baby breathe?

- Safe and appropriate methods of providing additional tactile stimulation include:
 - ✓ Slapping or flicking the soles of the feet
 - ✓ Gently rubbing the newborn's back, trunk, or extremities

Slapping the Sole of the Foot



Flicking the Heel



Rubbing the Infant's Back



Tactile Stimulation

Harmful Actions

- Slapping the back
- Squeezing the rib cage
- Holding upside down
and shaking,

Consequences

- Bruising
- Fractures, pneumothorax,
death
- Intraventricular bleeding
brain damage.

What do you do after the initial steps?

➤ Evaluate the baby in the following order:

1. **Respiration:** There should be good chest movements, and the rate and depth of respirations should increase after a few seconds of tactile stimulation.
2. **Heart rate:** The heart rate should be >100 bpm.
3. **Colour:** The baby should have pink lips and trunk. There should be no central cyanosis once the baby has good respiration and heart beat that indicates hypoxia.

What do you do if the baby is breathing, but has central cyanosis?

- The baby's skin color is best determined by looking at the central part of the body (**lips, tongue and central trunk**)
- If the baby is breathing but appears blue, administration of supplement oxygen is indicated.
 - ✓ Give free-flow oxygen
- Give a flow of 5 L/min with a tubing by cupped hand or a mask kept closer to the face.

Cupped hand



Mask held tightly

Free flow oxygen delivery methods

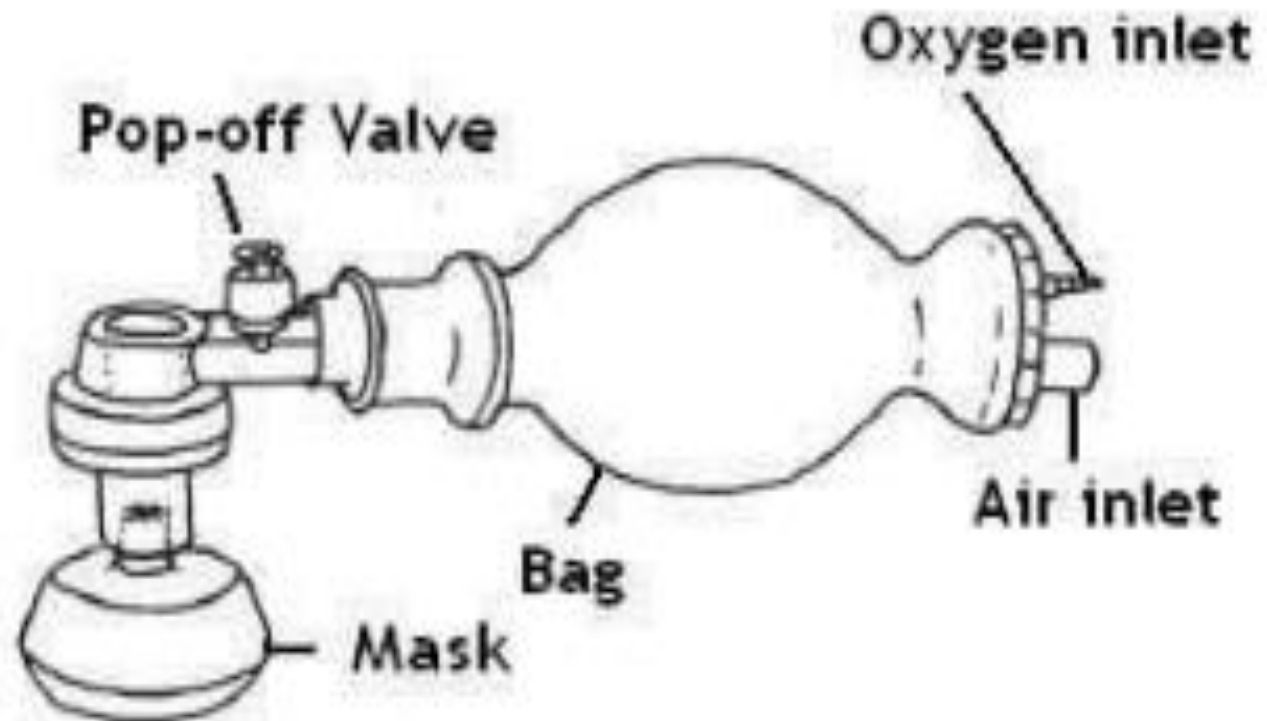
How do you know when to stop giving oxygen?

- When the newborn no longer has central cyanosis, gradually withdraw the supplemental oxygen until the newborn can remain pink while breathing room air.
- If cyanosis persists despite administration of free-flow oxygen, the baby may have significant lung disease, and a trial of positive-pressure ventilation may be indicated.

Observational Care

- Babies improving after initial steps of resuscitation require observational care.
- This can be provided either with the mother or in a newborn care unit.
- Assess breathing, heart rate, color and temperature every 15 minutes during the first hour after birth.
- Continue to provide warmth and initiate breast feeding.
- If baby is unstable, he will require additional interventions.

Use of Self Inflating bag to ventilate newborns



Appropriately sized masks

- ❖ A variety of mask sizes, appropriate for babies of different sizes, should be available at every delivery, since it may be difficult to determine the appropriate size before birth.
- ❖ The mask should cover the **chin, mouth, and nose**, but not the eyes, while still being small enough to create a tight seal on the face.

What characteristics of face masks make them most effective for ventilating newborns?

- Masks come in a variety of shapes, sizes and materials.
- Selection of a mask for use with a particular newborn will depend on how well the mask fits the newborn's face.
- The correct mask will achieve a tight seal between the mask and the newborn's face.
- For the mask to be of the correct size, the rim will cover the tip of the chin, the mouth, and the nose but not the eyes.

Cont'd...

- Too large-may cause possible eye damage and will not seal well
- Too small-will not cover the mouth and nose and may occlude the nose.
- Be sure to have various-sized masks available.
- Effective ventilation of a preterm baby with a term-infant size mask is impossible.

1



2



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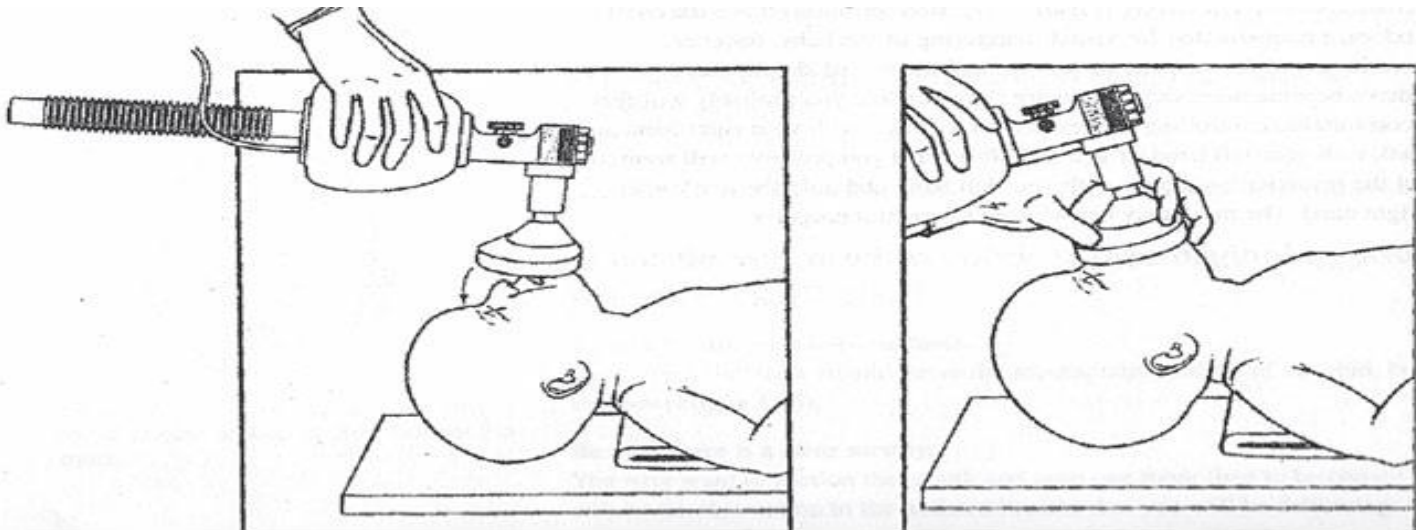
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1. Correct size and position
2. Mask too large, overlaps the chin
3. Mask too small, nostrils not covered
4. Mask too large, overlap with eyes

❖ How do you position the bag and mask on the face?

- You may find it helpful to begin by cupping the chin in the mask and then covering the nose.
- The mask usually is held on the face with the thumb, index, and/or middle finger encircling much of the rim of the mask, while the ring and fifth bring the chin forward to maintain a patent airway.



How often should you squeeze the bag?

- During the initial stages of neonatal resuscitation, breaths should be delivered at a rate of 40 to 60 breaths per minute, or slightly less than once a second.
- To help maintain a rate of 40 to 60 breaths per minute, try saying to yourself as you ventilate the newborn:
- Breathe....Two...Three.....Breathe.....Two.....Three...
- (squeeze) (release.....) (squeeze) (release.....)

How do you know if the baby is improving and that you can stop positive pressure ventilation?

❖ Improvement is indicated by the following 4 signs:

1. Increasing heart rate
2. Improving color
3. Spontaneous breathing
4. Improving muscle tone

What do you do if the heart rate, color, and muscle tone do not improve and baby's chest is not moving during positive-pressure ventilation?

- It may be due to one or more of the following reasons.
 1. The seal is inadequate
 - ✓ The most common place for a leak to occur is between the cheek and bridge of the nose.
 2. The airway is blocked
 3. Not enough pressure is being given

Chest Compression

- ❖ **What are the indications for beginning chest compressions?**
- Chest compressions should be started whenever the heart rate remains less than 60 bpm despite 30 seconds of effective positive-pressure ventilation(Check HR for 6 seconds)

What are chest compressions?

- Chest compressions, sometimes referred to as external cardiac massage, consist of rhythmic compressions of the sternum that
 - ✓ Compress the heart against the spine
 - ✓ Increase the intrathoracic pressure
 - ✓ Circulate blood to the vital organs of the body

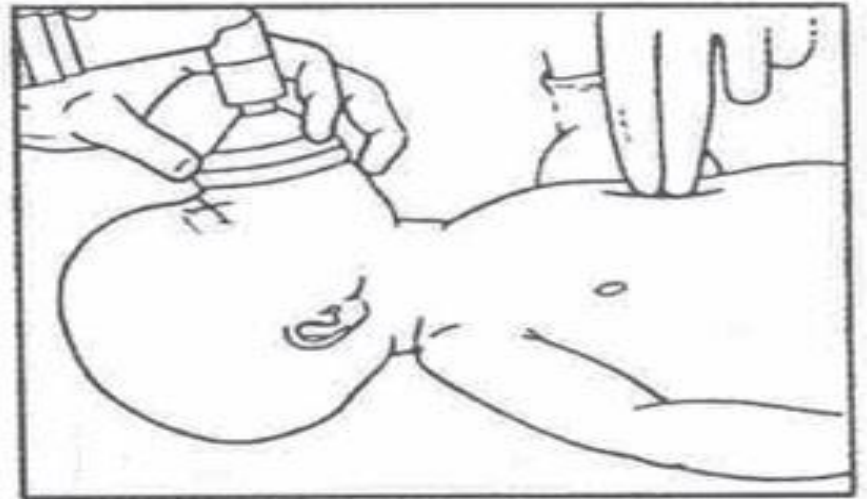
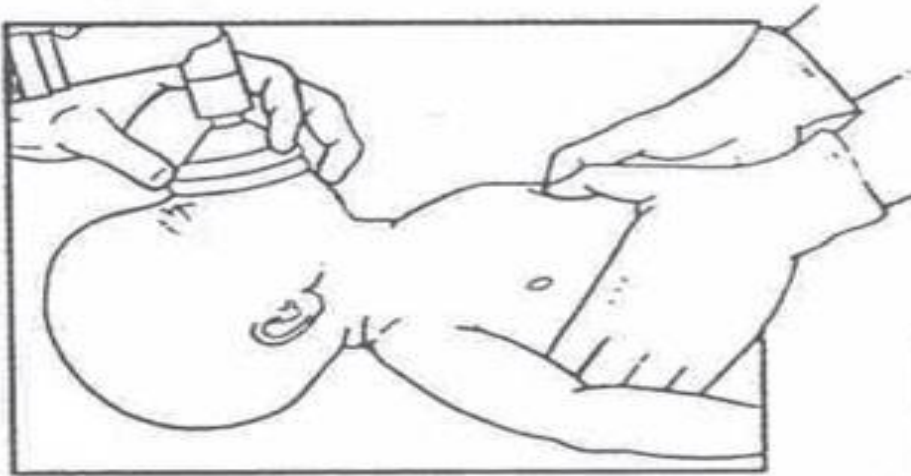
How many people are needed to administer chest compressions, and where should they stand?

- Remember that chest compressions are of little value unless the lungs are also being ventilated with oxygen.
- Therefore, 2 people are required to administer effective chest compressions one to compress the chest and one to continue ventilation.

How do you position your hands on the chest to begin chest compressions?

- There are two techniques for performing chest compression.
- 1. **Thumb technique**: where the 2 thumbs are used to depress the sternum, while the hands encircle the torso and the fingers support the spine.
- 2. **2-finger technique**: where the tips of the middle finger and either the index finger or ring finger of one hand are used to compress the sternum, while the other hand is used to support the baby's back (unless the baby is on a very firm surface).

Thumb technique



2-finger technique

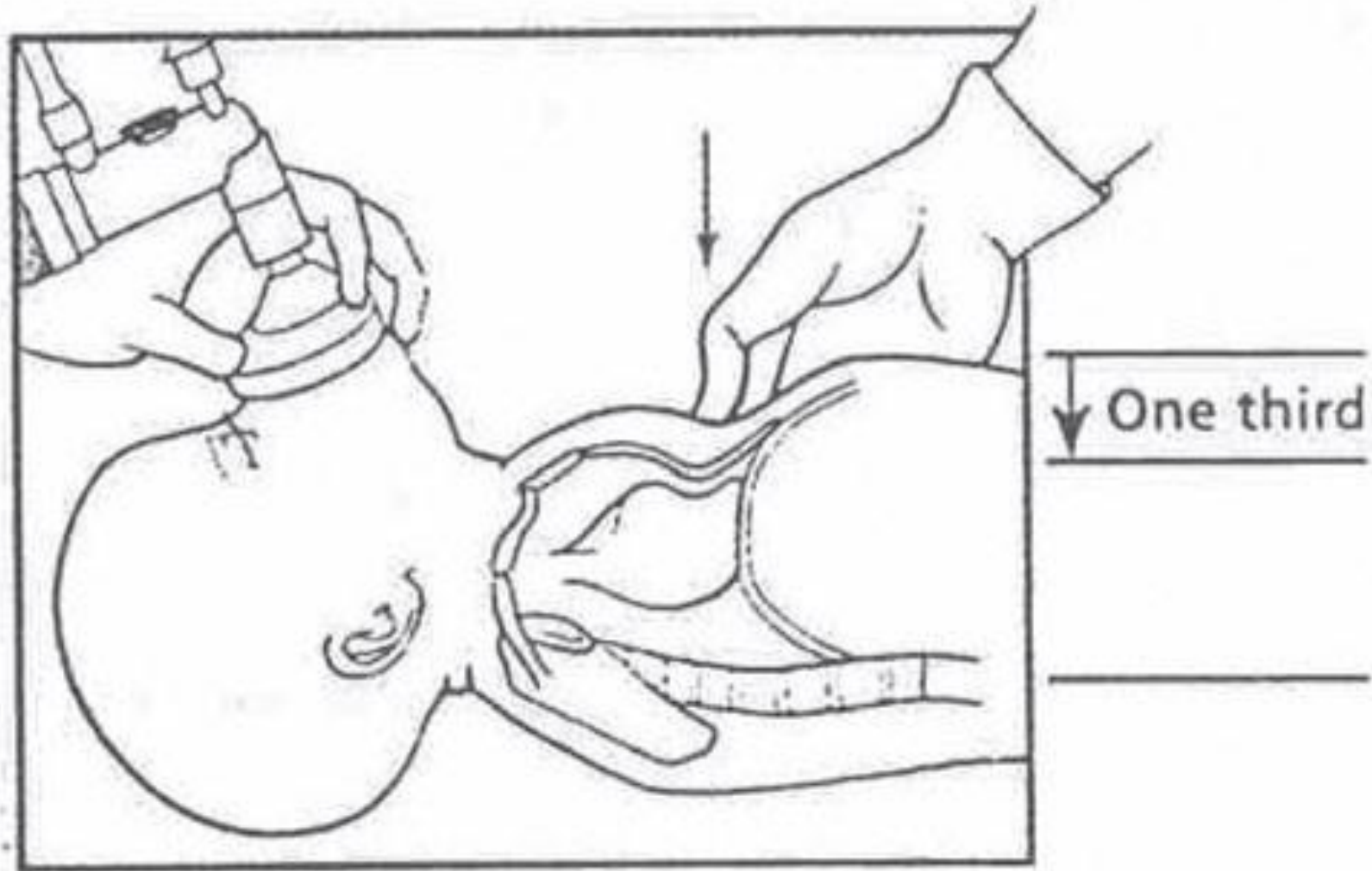
What are the advantages of one technique over the other?

- The **thumb technique** is preferred because it usually is less tiring, and you can generally control the depth of compression.
- This technique is superior in generating peak systolic and coronary perfusion pressure.
- However, the 2-finger technique is more convenient if the baby is large or your hands are small.
- The 2-finger technique also is preferable to provide access to the umbilicus when medications need to be given by the umbilical route.

How much pressure do you use to compress the chest?

- Controlling the pressure used in compressing the sternum is an important part of the procedure.
- One third of the anterior posterior diameter of the chest.
- One compression consists of the downward stroke plus the release.
- The actual distance compressed will depend on the size of the baby.

Do not lift the thumb or fingers completely off the sternum after each compression



Are there dangers associated with administering chest compressions?

- Chest compressions can cause trauma to the baby.
- Two vital organs lie within the ribcage (the heart and lungs).
- Pressure applied too low, over the xiphoid, can cause laceration of the liver.
- Also, the ribs are fragile and can easily be broken.

How often do you compress the chest and coordinate compressions with ventilation?

- chest compressions must always be accompanied by positive-pressure ventilation.
- Avoid giving a compression and ventilation simultaneously.
- The 2 activities must be coordinated, with one ventilation interposed after every third compression.
- For a total of 30 breaths and 90 compressions per minute.

How can you practice the rhythm of chest compressions with ventilation?

- ❖ Practice saying the words and compressing the chest.
 - One-and-Two-and-Three-and-Breathe-and-One-and-Two-and-Three-and-Breathe-and-One-and-Two-and-Three-and-Breathe-and-One-and-Two-and-Three-and-Breathe-and
- ❖ Now time yourself to see if you can say and do these five events in 10 seconds. Remember, squeeze your hand only when you say “Breathe and”

When do you stop chest compressions?

- After approximately 30 sec of well-coordinated chest compressions and ventilation, stop compressions long enough to determine the heart rate again. Feel the pulse at the base of the cord,
- **If the heart rate is >60 bpm, then** Discontinue chest compressions, but continue PPV now at a more rapid rate of 40 to 60 bpm.
- Once the heart rate rises above 100 bpm and the baby begins to breathe spontaneously, slowly withdraw PPV and move the baby to the nursery for post-resuscitation care.

? If the heart rate is now below 60 bpm, then

- Despite good ventilation of the lungs with positive-pressure ventilation and improved cardiac output from chest compressions,
- A small number of newborns (fewer than 2 per 1,000 births) will still have a heart rate below 60 bpm.
- **These babies would require epinephrine to stimulate the heart.**

How should you prepare epinephrine, and how much should you give?

- Although epinephrine is available in both 1:1,000 and 1:10,000 concentrations, the 1:10,000 concentration is recommended for newborns.
- Give 0.1 – 0.3ml/kg **IV** of 1:10,000 solution.
- Calculate the dose and give in 1ml syringe intravenously.

What should you expect to happen after giving epinephrine?

- Check the baby's heart rate 30 seconds after administering epinephrine.
- As you continue PPV and chest compressions, the heart rate should increase to more than 60 bpm after you give epinephrine.
- If this does not happen, you can repeat the dose after 3 to 5 minutes.

What should you do if the baby is in shock, there is evidence of blood loss, and the baby is responding poorly to resuscitation?

- Babies in shock appear pale, have delayed capillary refill and have weak pulses.
- They may have a persistently low heart rate, and circulatory status often does not improve in response to effective ventilation, chest compressions, and epinephrine.
- **Administration of a volume expander may be indicated.**

What can you give to expand blood volume? How much should you give? How can you give it?

- The recommended solution for acutely treating hypovolemia is an isotonic crystalloid solution.
- Acceptable solutions include
 - 0.9% NaCl (“Normal saline”)
 - Ringer’s lactate
- The initial dose is 10ml/kg. However, if the baby shows minimal improvement after the first dose, you may need to give another dose of 10ml/kg.
- In unusual cases of large blood loss additional dose might be considered.
- A volume expander must be given into the vascular system.

When to terminate resuscitation?

- After 10 minutes of continuous and adequate efforts if there are no signs of life (no heart rate and no respiratory effort), discontinue resuscitative efforts.
- Remember in all so-called ‘stillbirths’ the resuscitation efforts must be continued for 10 min
- The data suggests that in fresh stillbirths prognosis is not all that bad.

Thank You